

Logistic Regression

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1 Introduction

In statistics, the logistic model (or logit model) is a statistical model that models the log-odds of an event as a linear combination of one or more independent variables. In regression analysis, logistic regression (or logit regression) is estimating the parameters of a logistic model (the coefficients in the linear or non linear combinations).

We can choose from three types of logistic regression, depending on the nature of the categorical response variable:

Binary Logistic Regression: Used when the response is binary (i.e., it has two possible outcomes). The cracking example given above would utilize binary logistic regression. Other examples of binary responses could include passing or failing a test, responding yes or no on a survey, and having high or low blood pressure.

Nominal Logistic Regression: Used when there are three or more categories with no natural ordering to the levels. Examples of nominal responses could include departments at a business (e.g., marketing, sales, HR), type of search engine used (e.g., Google, Yahoo!, MSN), and color (black, red, blue, orange).

Ordinal Logistic Regression: Used when there are three or more categories with a natural ordering to the levels, but the ranking of the levels do not necessarily mean the intervals between them are equal. Examples of ordinal responses could be how students rate the effectiveness of a college course (e.g., good, medium, poor), levels of flavors for hot wings, and medical condition (e.g., good, stable, serious, critical).

2 Assumptions

To apply the Logistic Regression, the following assumptions are used:

- **The observations are independent.** It is assumed that the observations in the dataset are independent of each other. That is, the observations should not come from repeated measurements of the same individual or be related to each other in any way.

- **There is no severe multi-col-linearity among predictor variables.** It is assumed that none of the predictor variables are highly correlated with each other.
- **There are no extreme outliers.** It is assumed that there are no extreme outliers or influential observations in the dataset.
- **There is a linear relationship between the predictor variables and the logit of the response variable.** This assumption can be tested using a *Box-Tidwell Test*.
- **The sample size is sufficiently large.** As a rule of thumb, you should have a minimum of 10 cases with the least frequent outcome for each explanatory variable. For example, if you have 3 explanatory variables and the expected probability of the least frequent outcome is 0.20, then you should have a sample size of at least $(10*3) / 0.20 = 150$.

In addition, corresponding to the three types of logistic regressions, we have different requirements for the outcome respectively:

- **For binary logistic regression:** The dependent variable should have only two possible outcomes or categories. This can be verified by inspecting the dependent variable to ensure it has only two categories.
- **For multinomial logistic regression:** The dependent variable should have three or more unordered categories or outcomes. This can be verified by inspecting the dependent variable to ensure it comprises multiple unordered categories.
- **For ordinal logistic regression:** The dependent variable should have three or more ordered categories or outcomes, with a natural ranking among them. This can be verified by inspecting the dependent variable to ensure it consists of multiple ordered categories with an inherent hierarchy.

3 Algorithm

3.1 Construction

The general formula for Logistic Regression is given by:

$$\ln \left(\frac{P(X)}{1 - P(X)} \right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n = \beta_0 + \sum_{j=1}^n \beta_j X_j \quad (1)$$

Where:

X_j denotes the j^{th} predictor variable and β_j denotes the coefficient estimate for the j^{th} predictor variable.

Following Eq.1, we can obtain the function for the probability $P(X)$:

$$P(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n}}{1 + e^{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n}}$$

From which we can further derive the likelihood function for logistic regression:

$$L(\beta; \mathbf{y}, \mathbf{X}) = \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i} = \prod_{i=1}^n \left(\frac{e^{\beta X_i}}{1 + e^{\beta X_i}} \right)^{y_i} \left(\frac{1}{1 + e^{\beta X_i}} \right)^{1-y_i}$$

And yields the log likelihood:

$$\mathcal{L}(\beta) = \sum_{i=1}^n [y_i \ln(\pi_i) + (1 - y_i) \ln(1 - \pi_i)] = \sum_{i=1}^n [y_i \beta X_i - \ln(1 + e^{\beta X_i})]$$

As we always aim to maximise the likelihood function (or the log likelihood) in regression models, taking first derivative of the log likelihood and solving for $\mathcal{L}'(\beta) = 0$ gives the optimal value of β .

$$\mathcal{L}'(\beta) = \frac{\partial}{\partial \beta} \sum_{i=1}^n [y_i X_i \beta - \ln(1 + e^{X_i \beta})] = \sum_{i=1}^n \left(y_i X_i - \frac{X_i e^{\beta X_i}}{e^{\beta X_i} + 1} \right) = 0$$

Therefore we have, for all integers n :

$$\beta = \sum_{i=1}^n \frac{\ln\left(\frac{y_i}{1-y_i}\right) + 2i\pi n}{X_i}$$

As we can see, there doesn't exist a closed-form solution, hence we apply methods like the iteratively re-weighted least squares to find an estimate of the regression coefficients, $\hat{\beta}$.

The logistic regression can be applied using the R software (https://rpubs.com/mbounthavong/logistic_regression).

3.2 Dummy Variables

In most cases, the variables in a linear regression model is numeric. However, sometimes we may face categorical variables (with only several values to take). That's when we introduce the dummy variable to deal with them.

Consider the blood type variable with four values (A, B, AB, O) to take. To include such variables into the regression model, we conduct the following transformation:

Blood type	x_1	x_2	x_3
O	0	0	0
A	1	0	0
B	0	1	0
AB	0	0	1

Table 1: An example of using dummy variables

We break down the blood type variable into three separate ones. Setting blood type O as baseline, which all variables x_1, x_2, x_3 take values of 0. For blood type A, B and AB, only the corresponding variable from x_1, x_2, x_3 will take 1 and the rest remain 0. In this way, the influence of blood type O on the output will be expressed in the intercept while the coefficients for x_1, x_2, x_3 will be relative effect when blood type A, B and AB are compared with O.

For ordered categories, we often choose the lowest level to be the baseline level. In general, if the variable have n values to take, then $n - 1$ dummy variables are needed to fully include the variable into the model.

4 Akaike Information Criterion (AIC)

The Akaike information criterion (AIC) is a metric that is used to compare the fit of different regression models. In logistic regression, we often use it to determine the number of variables we need.

The AIC value is calculated as:

$$\text{AIC} = 2K - 2\mathcal{L}$$

Where K denotes the number of model parameters (variables) and $\mathcal{L}$ denotes the log likelihood of the regression model.

There's no fixed criteria for defining a "good" or a "bad" AIC value. However, the model with the lowest AIC offers the best fit. In addition, the absolute value of the AIC value is not important. For example, if Model 1 has an AIC value of 730.5 and Model 2 has an AIC value of 456.3, then Model 2 offers a better fit.

5 Hosmer–Lemeshow Test

The AIC value is a useful way to determine which regression model fits a dataset the best among a list of potential models, but it doesn't actually quantify how well the model fits the data.

For example, a particular regression model might have the lowest AIC value among a list of potential models, but it may still be a poor fitting model.

To determine if a model fits a dataset well, we can use the Hosmer–Lemeshow test. Its statistic is given by:

$$H = \sum_{g=1}^G \left(\frac{(O_{1g} - E_{1g})^2}{E_{1g}} + \frac{(O_{0g} - E_{0g})^2}{E_{0g}} \right) \quad (2)$$

We have $N_g = O_{1g} + O_{0g} = E_{1g} + E_{0g}$, and substituting E_{1g} with $N_g\pi_g$, we can obtain:

$$H = \sum_{g=1}^G \left(\frac{(O_{1g} - E_{1g})^2}{N_g\pi_g} + \frac{[(N_g - O_{1g}) - (N_g - E_{1g})]^2}{N_g(1 - \pi_g)} \right) = \sum_{g=1}^G \frac{(O_{1g} - E_{1g})^2}{N_g\pi_g(1 - \pi_g)}$$

The test statistic asymptotically follows a χ^2 distribution with $G - 2$ degrees of freedom. Since the null hypothesis H_0 of this test is that the regression model is of good fit to the data, we expect the p value to be larger than 0.05 so that we cannot reject the null hypothesis, that is, the regression model fits the data well.

The goodness of fit test can be applied in R software (see https://rpubs.com/mbounthavong/logistic_regression_part2).